CI HW02 Q01 Report

Problem Formulation

Ant Representation:

Each ant is represented as a list of routes where each route is itself a list. Each route represents the path a truck takes throughout the city. The first stop is the first place visited after leaving the depot while the last stop is the last place visited before returning to the depot.

Fitness Function:

The fitness function for each ant is calculated as the sum of the total distance that each truck in it will have to travel. The algorithm attempts to minimize the fitness function.

Hard Constraints:

1) Each stop can only be visited once
2) A truck cannot exceed its waste capacity
3) All routes start and end at the depot

Ant Construction:

Unvisited customers are stored as a set. The algorithm keeps constructing an ant until this set is empty. For each truck, the algorithm enters a while loop which keeps reconstructing a list of allowed stops the truck can visit without breaking any of the hard constraints. Once the allowed list for a truck is empty, the algorithm breaks the while loop and begins constructing the next truck. While constructing a truck, the next stop is chosen stochastically. Once a truck cannot visit any more bins, its route is added to the ant and the construction of the next truck route begins.

Pheromone Representation and Updating (tau):

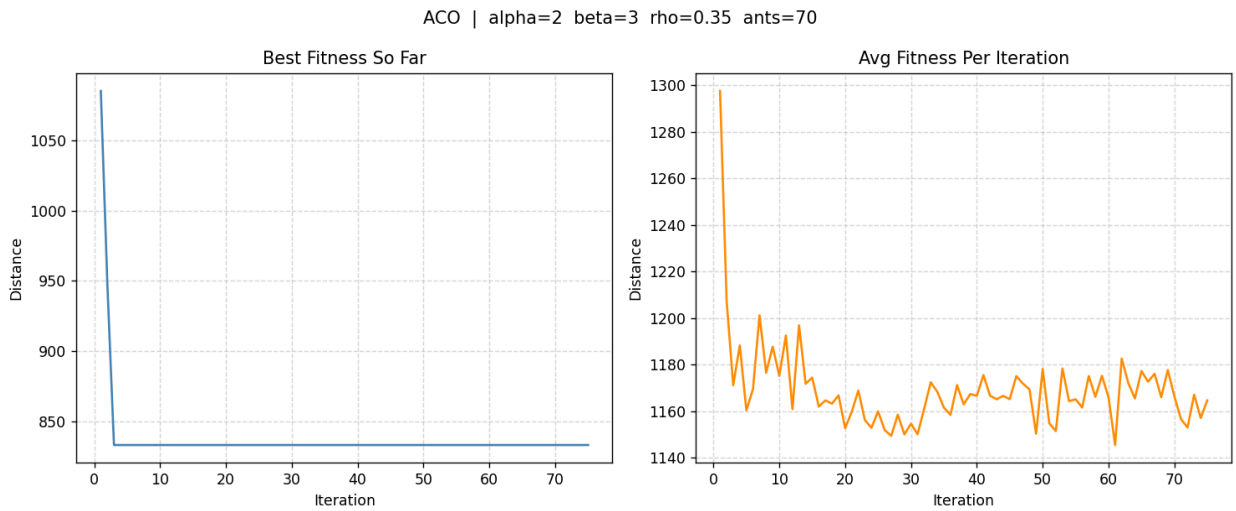
Pheromones represent the desirability of visiting stop j from stop i. The pheromone deposit for each ant is calculated using Q/fitness, where Q is a constant and fitness is calculated using the aforementioned fitness function. The pheromone deposit for an ant is then spread across all edges in all its routes. Pheromones are also evaporated.

Heuristic (eta):

1/distance is used as the eta for all edges.

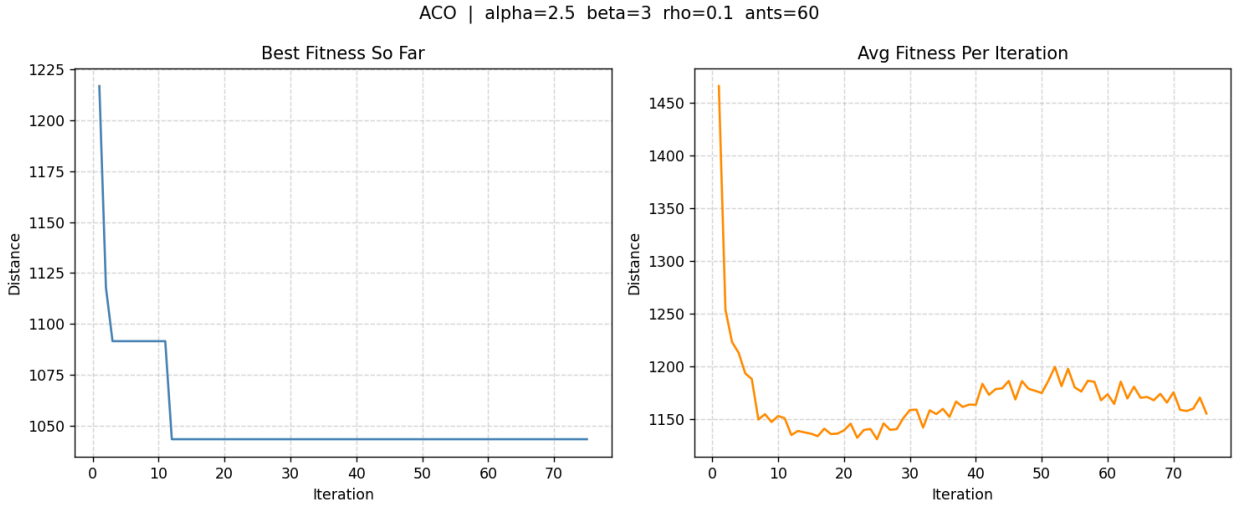
Test Case Results

Test Case 1:



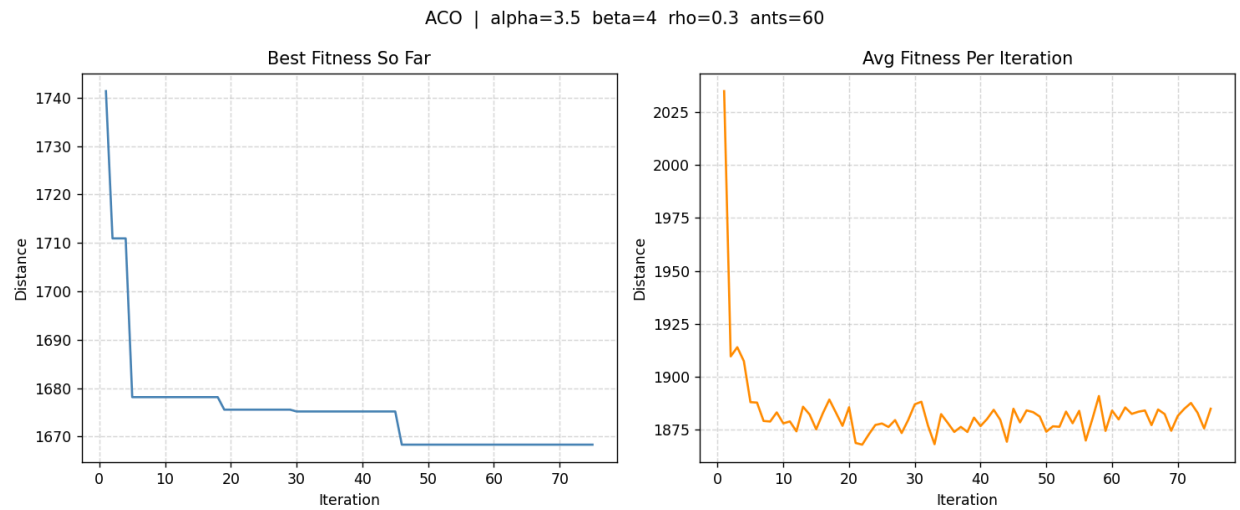
Best distance: 832.99

Test Case 2:



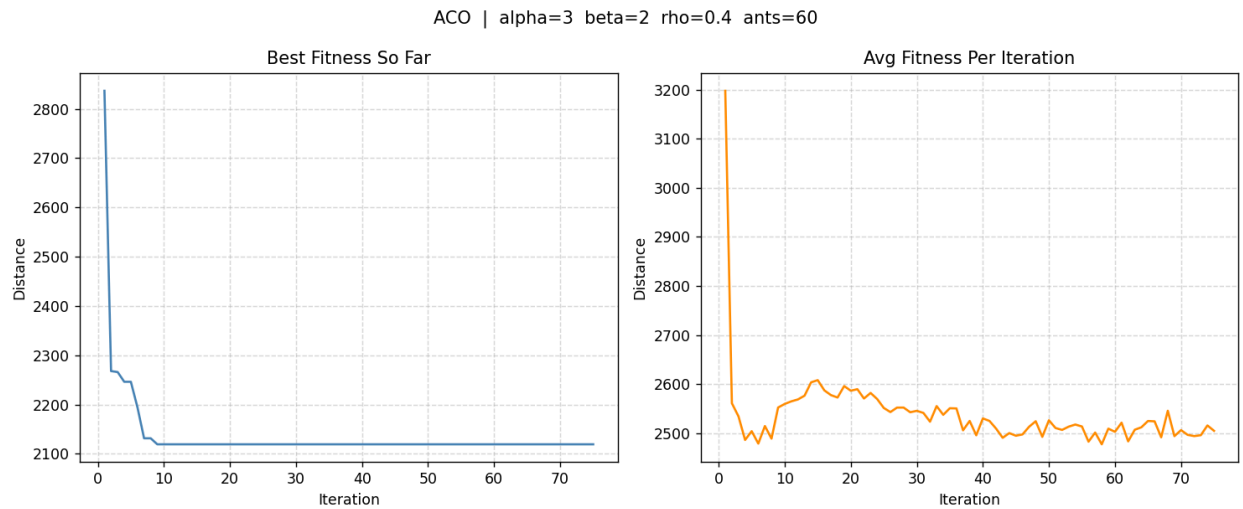
Best distance: 1043.40

Test Case 3:



Best distance: 1668.32

Test Case 4:



Best distance: 2119.30